

An Introduction to

Cyber Security

A Beginner's Guide



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Table of Contents

Introduction	01
An overview of Cyber Security	02
Threat examples of Cyber Security	03
The threats challenge of Cyber Security	04
Industries that have a high risk of Cyber Security	05
A quick glossary of Cyber Security terminologies	07
How to building a career in Cyber Security	11
Get ready to launch your career in Cyber Security	12
Conclusion	12

Introduction

The belief that “it will not affect us” is the biggest blunder that a modern-day organization can make on the issue of cybersecurity. The cybersecurity threat is real and it is now a worldwide problem.

In this digital era, every organization be it SMEs or large corporations, governments or banks, faces the threat of a system hack, ransomware attack, data breach or malware. Cyber-criminals of today are not old-time lone hackers. They run organized crime networks and often operate like startup companies, hiring highly-trained programmers to innovate new online attacks.

With organizations having to secure an increasing amount of sensitive data, cybersecurity is becoming relevant and essential for businesses of all sizes. As the scale and scope of threats continue to rise at a fast pace, fresh opportunities are opening up for qualified data professionals in diverse sectors.

Experts believe that the proliferation of IoT-connected devices, cloud-based applications, a range of technology initiatives and strict privacy protection mandates are driving the rapid growth of the cybersecurity market.

Cybersecurity is undoubtedly one of the fastest-growing fields. However, there is an acute shortage of skilled workforce. If you are enthusiastic about securing organizational digital assets but are unsure if a cybersecurity career is right for you, the basics of cyber security handbooks will clear your doubts.

Starting with cybersecurity introduction, this cybersecurity handbook will discuss everything you should know about cybersecurity threats, the worst affected sectors, security analytics use cases, cybersecurity terminologies and the skills in demand.

An overview of Cyber Security

Cybersecurity is a set of processes, technologies and methods to protect servers, computers, networks, electronic systems, data and mobile devices from unauthorized access through malicious attacks.

Securing the availability, confidentiality and integrity of an organization's digital assets and software against internal or external threats is the primary objective of cybersecurity.

The term cybersecurity encompasses a wide variety of security solutions. Common cybersecurity categories include-

- **Application Security:** Compromised applications can facilitate unauthorized access to data. Application security protects devices, applications and software from cyber-attacks.
- **Operational Security:** From access permissions to decisions and processes that determine where and how data will be shared or stored and fall under operational security. It places a great emphasis on securing data assets.
- **Information Security:** All about protecting the privacy and integrity of data in transit and in storage.
- **Network Security:** The process of ensuring security to computer networks from opportunistic malware or targeted attackers.
- **End-user Education:** This cybersecurity category covers an extremely unpredictable factor-humans. End-user education aims to teach users about cybersecurity threats and the best security practices to avoid them.

Cybersecurity is essential in protecting digital systems, networks and data from cyber threats such as hacking, malware, phishing and data breaches. It involves various practices, including encryption, firewalls, intrusion detection and user awareness to ensure confidentiality, integrity and availability of information. As technology advances, cybersecurity continues to evolve, requiring constant vigilance, proactive defense strategies and strong policies to safeguard individuals and organizations from ever-growing cyber risks.

Threat examples of Cyber Security

Cybersecurity solutions protect against three types of cyber threats, which are-

- **Cyber Attack:** Unauthorized information gathering.
- **Cyber Crime:** Groups or single actors targeting systems, networks, or servers for monetary benefit or for causing disruption.
- **Cyber Terrorism:** Undermines electronic system to cause fear or panic.
- **Data Espionage:** State-sponsored or corporate spying to steal confidential information.
- **Hactivism:** Cyber-attacks driven by political or social activism.
- **Insider Threats:** Employees or trusted individuals intentionally or accidentally compromising security.
- **Supply Chain Attacks:** Targeting third-party vendors to infiltrate larger organizations.
- **AI-Powered Cyber Threats:** Using artificial intelligence for automated hacking, deepfake fraud, or bypassing security measures.
- **Disinformation Campaigns:** Spreading false information to manipulate public opinion or destabilize societies.
- **Cloud Security Breaches:** Exploiting vulnerabilities in cloud infrastructure to access sensitive data.
- **IoT Exploits:** Attacks targeting smart devices, leading to unauthorized surveillance or system takeovers.
- **Cyber Insurance:** Financial protection policies against losses from cyberattacks and data breaches.

Cybersecurity threat is an ongoing challenge for individuals, businesses and governments due to the constant evolution of these threats. These threats highlight the importance of continuous cybersecurity measures to safeguard sensitive data and systems.

The threats challenge of Cyber Security

The methods used by cyber attackers, criminals or cyber terrorists to gain control over an organization's computer system include-

- **Malware:** Malware is malicious software. Hackers or cyber criminals create malware to damage or disrupt computer systems. Criminals also use malware to launch cyberattacks or to extract money.

Viruses, Spyware, Trojans, Adware, Botnets, and Ransomware, are common instances of malware.

- **Phishing:** Phishing targets its victims through authentic-looking bank or company emails, asking for confidential data, such as personal information or credit card details.
- **Man-in-the-Middle Attack:** In a man-in-the-middle attack, internet criminals intercept communication between 2 individuals for stealing data. Cybercriminals carry out these attacks on unsecured networks.
- **Denial-of-Service Attack:** This attack overwhelms servers and networks with traffic overload, denying computer systems from completing legitimate requests.
- **SQL Injection:** Cybercriminals use a Structured Language Query (SQL) injection to gain control over databases. This gives attackers access to sensitive data stored in databases.
- **Zero-Day Exploits:** Cybercriminals take advantage of unknown vulnerabilities in software before developers can release a fix, making these attacks extremely dangerous.
- **Credential Stuffing:** Attackers use leaked username-password combinations from data breaches to gain unauthorized access to user accounts across different platforms.

Cybersecurity continues to employ a variety of strategies to steal data, disrupt systems and exploit vulnerabilities, including malware, phishing, man-in-the-middle attacks and ransomware. Understanding these threats is crucial for implementing strong security measures and protecting sensitive information in the digital age.

Industries that have a high risk of Cyber Security

The Theft Resource Center identified many data breaches across different industries and left million personal data or records at risk. And security pros detected several hacking incidents, and now it is no secret that the following industries are more vulnerable to security risks.

- **Health Care:** From AI to VR, voice search to the blockchain and chatbots, the healthcare industry is introducing innovative technologies to boost patient care. These advanced technologies improve the quality of life but unfortunately, because of them, cybercriminals are increasingly targeting the sector. Hackers exploit the medical industry to obtain important patient data available in clinics and hospitals, such as patient bank details, social security numbers and personal addresses.
- **Manufacturing:** Thanks to the Internet of Things and cloud technologies, the manufacturing sector has become more efficient in communicating with customers and supply chains. However, connected technologies bring with them multiple access points that allow cybercriminals to steal factory and customer data.
- **Retail:** Retail faces a constant cyber threat because of the huge transaction volume that takes place within the sector every day. Internet scammers target the retail industry to pick credit card details, online account data, and other customers' personal information.
- **Finance:** Cyber criminals hacked into the multinational credit reporting organization Experian PLC, a brokerage company Scottrade and the American stock market index, Dow Jones. And reports of hacking into PayPal accounts, Australian banks and banks in Southeast Asia. For understandable reasons, financial institutions are at the forefront of cyber-attacks.
- **Use Cases of Security Analytics:** Security Analytics is a cybersecurity approach that collects and analyzes data from diverse network sources to create proactive protection measures. With this knowledge, security analytics solutions produce actionable insights by automating threat hunting. There are three use cases of security analytics, which include-

Real-Time Analytics: Real-time analytics analyzes alerts coming from real-time security systems such as SIEM (Security Information and Event Management). For example, it can detect dangerous Internet Protocol addresses in real-time based on known threat patterns and severity.

Rule-Based Analytics: Drawing intelligence feeds on existing and potential threats, rule-based analytics defines and applies rule-based approaches to identify and counter known attacks or bad actors.

Batch Analytics: Unlike real-time and rule-based, batch analytics analyzes unknown attacks. It uses data profiling and statistical models to identify threats. Batch analytics can also facilitate the visualization of security vulnerabilities.

- **Cyber Security Applications:** Combining the above three security analytics models, cybersecurity provides the following solutions.
- **Cloud Monitoring:** The cloud presents unique challenges when IT scales. Cybersecurity ensures tight supervision of the cloud infrastructure and cloud-based applications.
- **Behavior Analysis:** Users continuously interact with an IT infrastructure, and their behavior determines the failure or success of an organization's security. Cybersecurity analytics monitors users for unusual behavior to detect malicious activities, insider threats and compromised accounts.
- **Network Analysis:** Traffic moves out and comes into the network all the time. However, because of the high volume, maintaining visibility over every network transaction is not possible. Rule-based security analytics provides network cybersecurity by setting standards and criteria.
- **Data Extrusion Detection:** Data extrusion or data exfiltration refers to unauthorized data transfer from computers. The transfer can be via physical access or automated, through malware. Unsanctioned data movement can lead to data leakages and theft. Cybersecurity, using security analytics, can identify data exfiltration and prevent data theft.
- **Insider Threat Detection:** An insider threat, caused by a careless, malicious or ignorant user, can be as damaging as an external threat. Insider threats, sometimes, can wreck a network. Cybersecurity analytics can detect insider threats by analyzing user behavior, such as unusual login times, abnormal email usage and unauthorized access requests.
- **Threat Hunting:** Simply responding to threats puts organizations in a tight corner. Cybersecurity automates threat hunting, which enables continuous scanning for dwelling threats and potential breaches. This implements a comprehensive and proactive protection measure.

A quick glossary of Cyber Security terminologies

- **Access:** Gaining control over a system's information or knowledge.
- **Adversary:** An individual or a group with criminal intent.
- **Antivirus:** A software that protects a computer from a malicious attack.
- **Asset:** Documents, information, resources, and other data that are of great importance.
- **Backdoor:** Instant access to a system without the need to log in.
- **Botnet:** Compromised or infected devices on an internet-connected network.
- **BYOD (Bring Your Own Device):** Employees bringing and using their own devices, authorized by an organization.
- **Cloud Computing:** On-demand availability of system resources, typically computing power and data storage without a user's active or direct management.
- **Ciphertext:** A plaintext encrypted with an encryption algorithm.
- **Confidentiality:** The state of keeping something private or secret.
- **Cyber Essentials:** Industry-supported, Government-backed policies to help protect organizations against online threats.
- **Data Breach:** The unintentional or intentional release of private or secure information.
- **DoS (Denial of Service):** A cyber-attack that disrupts services by denying access to network resources or machines.
- **Exfiltration:** Transfer of data without consent.
- **Decode:** To convert an encoded message into comprehensible language.
- **Ethernet:** The architecture for interconnecting computer systems via a wired local area network.
- **EUD (End-user device):** A device capable of storing information, such as a PC, laptop, smartphone, tablet, hard drive, memory card or USB flash drive.
- **Exploit:** An attempt to breach secure networks to gain access to digital assets.
- **Forensics:** An application of analysis and investigation to gather evidence from computing devices.
- **Firewall:** A security system that controls and monitors outgoing and incoming network traffic in conformity with predetermined rules.
- **GDPR (General Data Protection Regulation):** A law that regulates how organizations protect the personal data of EU citizens.
- **Hacker:** A cybercriminal who uses electronic devices to obtain unauthorized data access.
- **Hashing:** An algorithm applied to data to validate that the information is not corrupted, tampered, or modified.
- **ISO 27001:** A globally recognized standard for risk management of information security as per the ISMS (Information Security Management System) procedures and policies.

- **ICT (Information and Communications Technology):** An extended term for IT, stresses unified communications such as the integration of computers, mobile phones, wireless networks, the internet, middleware, enterprise software and media applications.
- **Integrity:** This refers to data that has not been tampered or modified.
- **Jailbreak:** A process of removing device security restrictions, allowing its user to make modifications.
- **Keystroke Logging:** Also called keylogging, it refers to capturing a computer user's keystrokes on a keyboard.
- **Logic Bomb:** A code inserted to set off malicious functions in software systems.
- **Macro Virus:** A virus programmed in the macro language. Microsoft Excel and Word are common applications featuring macro languages.
- **Malware:** Malicious software designed for disrupting, damaging, or gaining unauthorized access.
- **Network/Edge Security:** Connected computers linked through the internet. And Securing data processing and storage at the network's edge (closer to users and devices).
- **NIS Directive:** A European Union directive on the security of information systems and networks.
- **NIST Cybersecurity Standard:** National Institute of Standards and Technology (NIST) guidelines for US private enterprises to follow for improved detection and response to cyber-attacks.
- **Penetration Testing:** Often referred to as ethical hacking, it is a simulated cyber-attack to assess a system's security.
- **Phishing:** A fraudulent and criminal attempt to obtain sensitive information by disguising as a legitimate entity.
- **Quadrant:** Technology for making tamper-proof cryptographic equipment.
- **Outsider Threat:** An external security threat by a group or an individual.
- **Ransomware:** Malicious software that disrupts a system and demands a ransom to make it workable again.
- **SaaS (Software as a Service):** A software delivery model on the cloud on a subscription basis.
- **Security Perimeter:** A boundary between networks with necessary safeguards against cyber-attacks.
- **Steganography:** A technique used to conceal data within ordinary messages and files to bypass detection.
- **Two-Factor/Multi-Factor Authentication:** An authentication mechanism that grants system access to a user only after presenting two authorization evidence.
- **Unsigned Data:** A data type that holds positive values.
- **Virus:** A malicious program that replicates on infected computers.
- **Worm:** A malware type that spreads its copies from one computer to another.
- **Zero-Day:** An unaddressed or unknown software vulnerability.

- **Session Hijacking:** Attackers steal a user's session ID (like a cookie) to impersonate them and gain unauthorized access to web applications.
 - **Rootkits:** A type of malware that gives hackers deep control over a system, allowing them to hide malicious processes, access sensitive files, and manipulate system functions.
 - **Cryptojacking:** Cybercriminals hijack a victim's computer or network resources to mine cryptocurrency without their consent.
 - **SIM Swapping:** Attackers trick mobile carriers into transferring a victim's phone number to their own SIM card, gaining access to calls, texts and two-factor authentication codes.
 - **Drive-By Downloads:** Malicious software is automatically downloaded onto a user's device without their knowledge when visiting a compromised or malicious website.
 - **Watering Hole Attacks:** Hackers target websites frequently visited by specific individuals or organizations and inject malware to infect their devices.
 - **DNS Spoofing:** manipulate DNS records, redirecting users to fraudulent websites to steal credentials or install malware.
 - **Deepfake Attacks:** AI-generated fake videos or audio used for deception.
 - **Rogue AI Attacks:** Cybercriminals exploiting artificial intelligence for automated cyberattacks.
 - **Cyber Physical Attacks:** Attacks targeting both digital and physical infrastructure, such as power grids or smart cities.
 - **Cyber Hygiene:** Best practices for maintaining security, such as strong passwords and regular software updates.
 - **Firmware Attacks:** Targeting low-level system software (BIOS, UEFI) to gain deep access to a device.
 - **Air-Gap Attack:** A method of hacking isolated systems that are not connected to the internet.
 - **Dark Web Monitoring:** Tracking stolen credentials and cyber threats on hidden parts of the internet.
 - **Quantum Cryptography:** Security techniques leveraging quantum mechanics for ultra-secure communication.
 - **API Security:** Protecting APIs from threats like data leaks, unauthorized access and injection attacks.
 - **Shadow IT:** Unauthorized use of IT systems, applications or services by employees.
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- **Biometric Spoofing:** Using fake fingerprints, facial images or voice recordings to bypass biometric authentication.
 - **Synthetic Identity Fraud:** Creating fake identities using real and fabricated data for cyber fraud.

- **Data Poisoning:** Corrupting AI or machine learning datasets to cause incorrect predictions.
- **Embedded System Security:** Protecting firmware and software in hardware devices.
- **Synthetic Media Attacks:** Manipulating or generating media (audio, video, images) for misinformation or fraud.
- **Digital Twin Security:** Protecting virtual replicas of physical systems used in industries like healthcare and manufacturing.
- **Malvertising:** Malicious advertisements that deliver malware when displayed or clicked.
- **Fileless Malware:** Malware that operates in memory without leaving traces on the disk, making detection difficult.
- **Zero-Click Attacks:** These attacks exploit software vulnerabilities, often in messaging apps, email clients or operating systems to execute malicious code automatically do not require any interaction from the victim such as clicking a link or downloading a file
- **Digital Footprinting:** The collection and analysis of personal or organizational data available online to assess exposure risks.
- **MFA Fatigue Attack:** Repeatedly sending authentication requests to trick users into approving a malicious login.
- **Baiting Attack:** A form of social engineering where attackers leave malware-infected USB drives or links to lure victims.

How to building a career in Cyber Security

Cybersecurity skills to land a job to pursue a career in cybersecurity, you must have a range of skills, including technical, operational, management and communications. Here is a list of the basic skills you will need to enter the world of cyber defense.

Problem Solving:

Solving critical problems will be part of your daily work. You will need to be innovative in creating effective measures to address complex security challenges.

Technical Aptitude:

As a cybersecurity professional, you need to be technologically savvy to keep up with current and emerging threats. You must keep in mind that the work of a cybersecurity expert is purely technology-focused.

Familiarity With Multiple Platforms:

Cybersecurity is not restricted to computers. It also protects wireless networks, cloud infrastructure, and handheld devices. So, being familiar with various platforms is essential.

Attention to Details:

To secure large organizations from cyber-attacks, you need to be alert and attentive to details. As a cybersecurity professional, you will need to monitor the network 24/7 and if a threat occurs, you have to identify it swiftly and come up with security solutions in real-time. Therefore, staying attentive is vital.

Communication Skills:

From Management to Chief Security Officer, vendors and the general workforce, cybersecurity professionals must interact with every stakeholder. Having powerful communication skills is critical to effectively communicate security policies, concerns and solutions to all.

Forensics Skills:

While cybersecurity and computer forensics are two unique areas, they are, however, interrelated. Knowledge of computer forensics is necessary to understand what went wrong if your defense mechanism does not work. It also helps recover compromised data.

Get ready to launch your career in Cyber Security

Cybersecurity is a growing field on the verge of an actual revolution and the focus on related technologies, ideas and practices are going to increase multifold within the next few years. Businesses are seeking cybersecurity enthusiasts professionals who don't just have the basic skills but can think differently, respond to problems in the business proactively and constantly

innovate. Despite having a bachelor's degree in computer science is a good start, specialized training on the latest practices, tools and techniques are critical to position yourself as a competent expert in cybersecurity capable of sustaining both the present and future enterprise challenges.

Conclusion

In today's digital world, cybersecurity is not only a concern for IT professionals but also a necessity for everyone. From understanding basic threats like phishing and malware to adopting proactive security measures such as multi-factor authentication and secure browsing, beginners must recognize that cybersecurity is an ongoing process not a one-time fix.

You can significantly reduce risks and safeguard both personal and professional data by establishing a solid foundation in cybersecurity awareness, practicing safe online habits, and staying up to date on new threats. As technology evolves, so do cyber threats, making it essential to learn, adapt and implement security best practices regularly.

Keep in mind that cybersecurity is about more than just defense, it is also about equipping yourself with the knowledge necessary to safely navigate the digital world. Stay alert, stay informed and most importantly, stay cyber-safe!